

Ajzal Ashraf Valiyakath

<https://ajumon2001.github.io> Edinburgh, UK +44 7748904522 a3005@hw.ac.uk [linkedin.com/in/ajzal-ashraf-valiyakath/](https://www.linkedin.com/in/ajzal-ashraf-valiyakath/)

Education

M.Sc. Robotics with Industrial Application,

Heriot Watt University, UK

September 2024 - September 2026

- Gaining comprehensive, hands-on experience in robotic system design, machine vision, and sensor integration, with strong knowledge of robotic kinematics, AI-driven control, and ROS 2. Well-versed in 3D printing, digital twin development, and modern simulation and automation tools, enabling the end-to-end design, build, and deployment of advanced robotic systems.

B.Eng. Electronics and Computer

University of Georgia, Tbilisi, Georgia

April 2020 – April 2024

- Specialized in Electronics and Semiconductor systems, with additional focus on practical web development. Achieved a GPA of 3.0/4.0, demonstrating strong academic performance across both hardware and software domains.

Project Experience

Immersive Calibration of a 6-DOF Robotic Arm through VR-Driven Digital Twin Interaction

Working on my Master's dissertation, focused on the design and fabrication of a fully autonomous 6-DOF 3D-printed robotic arm with a 1 kg payload capacity. The system integrates a NVIDIA Jetson Orin Nano, Raspberry Pi 5, and a depth camera to enable real-time perception, control, and vision-based feedback. A custom gripper and embedded control framework support adaptive manipulation and precise motion execution. Using Unreal Engine, I am developing an immersive VR-driven digital twin to achieve real-time calibration, kinematic optimization, and enhanced accuracy for low-cost robotic manipulators.

Design and Control of a 3D-Printed 6-DOF Robotic Manipulator

<https://github.com/ajumon2001/6DOF-Robot-Arm>

Designed and developed a fully 3D-printed 6-DOF robotic manipulator for precision tasks. Each joint was actuated using high-torque servo motors controlled via an Arduino micro-controller. Implemented forward and inverse kinematics to achieve accurate end-effector positioning and validated performance through trajectory planning, motion smoothness, and repeatability testing. This project emphasized mechanical design, embedded control, and algorithmic implementation for real-time robotic actuation.

Surgical Assistance Using a 6-DOF Robotic Arm (Interdisciplinary Project)

Leading the development of a surgical assistance platform using a precision-controlled 6-DOF robotic arm in collaboration with medical students. The system incorporates AI-driven motion planning, sensor data analysis, and secure communication protocols to ensure safety and precision. Additional features include a ROS 2-based control architecture, haptic teleoperation for remote tasks, and vision-based control using Vision-Language Models (VLM) to enhance situational awareness. A curiosity-driven algorithm is implemented to enable intelligent and adaptive behaviors in dynamic surgical environments.

Storytelling Robot Using LLM and Voice Control (HRI – Marty Project)

github.com/ajumon2001/HRI-Marty-Group-6

Developed a voice-driven HRI prototype for Marty the Robot, integrating speech-to-text, LLM-based response generation, and text-to-speech into a reliable real-time interaction loop. Built modular pipelines for audio I/O and robot control, and troubleshoot connectivity and runtime issues to improve repeatability and system stability in live demonstrations.

Experience

Konpanion (Robotic Systems Engineer Intern)

The National Robotarium, Edinburgh, UK

September 2025 - Present

- Working on developing soft-robotic well-being platforms, operating at the intersection of robotics, IoT, and embedded systems across both R&D and product engineering. I build and maintain real-time monitoring dashboards for connected devices, support repeatable testing and data collection, and assist with prototype bring-up, calibration checks, and system upgrades. I troubleshoot hardware/software issues in Linux-based workflows, document procedures and outcomes, and collaborate with cross-functional teams to ensure reliable integration across sensors, firmware, and applications, improving system uptime and deployment readiness.

Aquanzo (Robotic Technician Intern)

University of St. Andrews, St Andrews, UK

July 2025 - January 2026

- Contributed to marine and aquaculture robotics projects with a focus on Brine Shrimp (Artemia) culture and sustainable aquaculture systems. Worked on developing robotic imaging and monitoring platforms integrating automation, computer vision, and embedded hardware to enhance environmental data collection. The work supported the development of scalable, intelligent solutions for real-time biological monitoring in aquaculture environments. Data annotation using Roboflow for Dataset ops.

Student Representative: M.Sc Robotics
Heriot Watt University, Dept. of Engineering and Physical Science

September 2024 - Present

- Elected to represent the student body during the IET Accreditation Visit, a critical evaluation process ensuring academic programmes meet professional engineering standards. Acted as the liaison between students, faculty, and industry assessors, communicating student experiences, raising curriculum feedback, and contributing to the successful accreditation of the programme. This role demonstrated leadership, stakeholder engagement, and the ability to advocate for academic quality and continuous improvement.

Assistant Lecturer & Lab Technician (Internship)
University of Georgia, Physics Department, Tbilisi

October 2023 - February 2024

- Assisted in conducting laboratory sessions and supported teaching and research activities, ensuring accurate experimental setup and smooth execution of physics experiments in an academic setting.

Assistant Mechanic & Sales Representative
Al-Najah Group Auto Parts and Accessories CO. W.L, Kuwait

April 2019 - December 2019

- Assisted in vehicle diagnostics and mechanical repairs, maintained parts inventory, and provided customer-facing support in a fast-paced automotive retail environment. Developed hands-on technical skills while ensuring efficient workflow and high customer satisfaction.

Certificates

- CAD & Design:** SolidWorks (Udemy)
- Programming:** Python, Web Development (Udemy)
- Machine Learning & AI:** Machine Learning A–Z: AI, Python & R (Udemy)
- Robotics & Simulation:** ROS in C++ (Udemy)

Technical Skills

- CAD & Design:** SolidWorks, Fusion 360 on Robotic Manipulators, End-Effectors, Mechanical assemblies and Tolerance Design.
- Programming:** Embedded C/C++, Python, MATLAB, HTML/CSS, MySQL, basic JavaScript.
- Embedded Systems:** Arduino, ESP32, Raspberry Pi, STM32, PWM, ADC/DAC, firmware development, real-time scheduling, Sensor Integration and Firmware basics.
- Robotics & Simulation:** ROS1/ROS2, URDF, RViz, Gazebo Classic/Sim, Unity3D for digital twins, System Integration, Edge Deployment Exposure and Calibration Concepts
- AI/ML:** TensorFlow, PyTorch, scikit-learn, OpenCV, CNNs, Autoencoders, RL, edge deployment (Jetson Orin NX, OpenMV).
- Prototyping:** 3D printing (FDM/PLA), laser cutting, PCB soldering, tolerance testing, iterative design-validation.
- Communication Protocols:** MQTT, I2C, SPI, UART, Wi-Fi, BLE (ESP-NOW), basic IoT networking.
- Tools:** GitHub, VS Code, ROS2-Gazebo Bridge, MATLAB Simulink, OpenCV with CUDA, Linux (Developmental/runtime), Automation Basics and Documentation practises.
- Special Focus:** Real-time motion control, tactile sensing, HRI, closed-loop feedback, simulation-to-hardware transfer.

Societies & Team Involvement

- AstroWatt – Robotics & Electronics Team:** Contributing to the design and development of robotic and electronic subsystems for student aerospace initiatives, focusing on control systems, embedded integration, and sensor interfacing.
- RocketWorks – Avionics Department:** Working on avionics systems for student rocketry projects, involving sensor integration, communication protocols, and control electronics for flight systems.

Languages

English:	Fluent	Malayalam:	Native
German:	Intermediate	Georgian:	Intermediate
Arabic:	Basic	Hindi:	Basic

Hobbies and Interests

- Photography and Machine Vision Calibration
- Researching Emerging Technologies in Robotics, AI, and Human-Machine Interfaces
- 3D Printing, DIY Mechatronics, and Robot Building & Tuning